26900350 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEMAT301 - MATHEMATICAL MUSINGS BEYOND CLASSROOM

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** questions

Each question carries **2** marks

1. What is meant by a bell shaped distribution? [U] / [CO1]
2. Mention the director and the main character the film "A Beautiful Mind". [K] / [CO4]
3. Name the specific Jesuit educational center in Goa/Cochin that served as the primary headquarters for collecting regional scientific knowledge on the Malabar Coast. [K] / [CO5]
4. Define "game formalism" as described in the philosophy of mathematics. [U] / [CO2]
5. What is the "blind spot" identified in consistent formal mathematical systems by Gödel's incompleteness theorems? [E] / [CO5]
6. Where was zero placed on the number line after it became a number? [K] / [CO6]
7. Write the name of two films that depict brilliant mathematicians serving crucial roles during major global military crises—one during World War II and the other during the Cold War Space Race? [K] / [CO1]

8. Name the Jesuit mathematician trained under Christopher Clavius who arrived in Goa in 1578 with specific instructions to investigate the sciences of India. [K] / [CO2]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Explain the realist response that treats the independence of the Continuum Hypothesis as a limitation of Zermelo–Fraenkel set theory with the Axiom of Choice(ZFC). [An] / [CO5]

10. Discuss the relationship between Ramanujan and G.H Hardy. [An] / [CO4]

11. Explain the history of unit fractions, detailing how Leonardo Fibonacci contributed to their expansion and the greedy procedure [An] / [CO5]

12. Explain how Mary Jackson overcame the educational barriers she faced to become an engineer [U] / [CO2]

13. Describe the series transformation technique used by Madhava to convert a slowly converging series into a rapidly converging series for an arc that is $1/12\text{th}$ of the circumference. [U] / [CO2]

14. Distinguish between ordinal numbers and cardinal numbers using von Neumann ordinals. [K] / [CO2]

15. Contrast the geometric technique of "direct rectification" (subdividing an arc into unequal parts) with the classical "method of exhaustion" [I] / [CO7]

16. What is PageRank? Explain how it measures the importance of a webpage. [U] / [CO6]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. Investigate how the film "The Imitation Game" depicts the multidisciplinary synthesis required at Bletchley Park, integrating abstract group theory, linguistics, statistics, engineering, and state espionage. [An] / [CO4]

18. State the historical dates of the Arabic and Latin translations of the Indian Siddhantas that established the westward transmission of astronomy. [K] / [CO5]

19. "The human brain is a finite object, yet mathematics is full of the infinite." Justify the statement. [U] / [CO2]

20. Detail Georg Cantor’s hierarchy of transfinite cardinalities and prove that the set of real numbers is uncountable ($2\aleph_0 > \aleph_o$). [U] / [CO2]

21.	Trace the development of number systems from early counting methods to the Babylonian system .	[U] / [CO6]
22.	Compare the cinematic representation of John Nash's mathematical creativity in A Beautiful Mind with the realities of academic mathematical discovery, discussing where Hollywood simplifies or dramatizes the research process.	[An] / [CO4]

[10x3 = 30]